MIL-C-43656A
3 December 1982
SUPERSEDING
MIL-C-43656
29 July 1969

MILITARY SPECIFICATION

CLOTH, COATED, POLYESTER, SILICONE RUBBER COATED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers the requirements for one type of silicone rubber coated polyester cloth (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-A-250/4	-	Aluminum Alloy 2024, Plate and Sheet
TT-S-735	-	Standard Test Fluids; Hydrocarbon
PPP-P-1136	-	Packaging and Packing of Coated (Plastic; Rubber) and Laminated Fabrics

MILITARY

MIL-H-5606	-	Hydraulic Fluid, Petroleum Base; Aircraft, Missile, and Ordnance
MIL-T-5624	-	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-L-6081	-	Lubricating Oil, Jet Engine

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods
FED-STD-406 - Plastics; Methods of Testing
FED-STD-601 - Rubber; Sampling and Testing

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1487 - Glossary of Cloth Coating Imperfections

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1424 - Tear Resistance of Woven Fabrics by Falling-Pondulum (Elmendorf) Apparatus

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 Samples.

* 3.1.1 Standard sample. The coated cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

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3.1.2 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.4).

3.2 Materials (see 6.6).

3.2.1 Base cloth. The base cloth shall be polyester sheeting conforming to the requirements of table 1, (see 4.4.1).

TABLE I. Physical requirements of base cloth

Characteristic	Requirement
Weight, ounces per square yard	3.3 ± 0.3
Weave	Plain
Color	Natural
Yarns per inch (minimum)	
Warp	50
Filling	50
Breaking strength, pounds (minimum)	
Warp	160
Filling	160

3.2.2 Coating compound. The coating compound shall be a composition of silicone rubber suitably compounded and pigmented to meet the requirements of this document (see 4.4.1).

3.3 Coated cloth. The coating compound specified in 3.2.2 shall thoroughly impregnate the base cloth, and it shall be applied equally to both sides to form a balanced coating. The coated cloth shall conform to the requirements of table II, 3.3.1 and 3.3.2. Testing shall be as specified in 4.4.3.

TABLE II. Physical requirements of coated cloth

Characteristic	Requirement
Weight, ounces per square yard	10.0 ± 0.5
Hydrostatic resistance, lbs./sq. in. (minimum)	
Initial	200
After strength of coating	200
After abrasion	100
After heat aging 1/	200
After crease resistance 2/	200

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TABLE II. Physical requirements of coated cloth (cont'd)

Characteristic	Requirement
Tearing strength, grams (minimum)	
Warp	4800
Filling	4800
Breaking strength, pounds (minimum)	
Warp	160
Filling	160
Adhesion of coating, lbs./2 inch width (minimum)	7.0
Stiffness, inches (maximum)	
Initial (warp only)	2.5
After heat aging (warp only)	2.5
At minus 85 ± 2°F (warp only)	3.0

1/ The coated cloth shall show no evidence of tackiness, flaking, stiffness or brittleness after completion of the heat aging part of the test specified in 4.4.3.

2/ The coated cloth shall show no evidence of cracking, flaking or separation after completion of the crease resistance part of the test specified in 4.4.3.

3.3.1 Fluid resistance. The coated cloth shall show no evidence of tackiness, peeling, stiffening or flaking when tested for fluid resistance as specified in 4.4.3.

3.3.2 Aluminum corrosion. The coated cloth shall not cause any pitting of aluminum alloy when tested as specified in 4.4.3.

3.4 Color. The color of the coated cloth shall be as specified in the contract or purchase order (see 6.2).

* 3.4.1 Matching. The color and shade shall match the standard sample (see 6.3) under artificial daylight having a color temperature of 7000 ± 500 kelvins and shall be a good approximation to the standard sample under incandescent lamplight at 2850 ± 100 kelvins.

3.5 Width. The minimum width of the coated cloth shall be as specified in the contract or purchase order.

* 3.6 Length and put-up. The coated cloth shall be put up in rolls as specified in PPP-P-1136 of not less than 100 yards. No roll shall contain more than three pieces and no individual piece shall be less than 10 yards. The ends of the pieces shall be overlapped and not joined by a seam.

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3.7 Odor. The coated cloth shall be free from any objectionable odor. Characteristic odor of coating is not considered objectionable.

3.8 Workmanship. The finished coated cloth shall conform to the quality of product established by this document. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.2.1 and tested for the characteristics specified in table V. The presence of any defect or failure of any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase documents. In addition, testing shall be performed for the characteristics specified in table I. All test reports shall contain the individual values used in determining the final result. The sample unit for the base cloth shall be 1 yard full width. The lot size shall be expressed in units of 1 yard and the sample size shall be as specified below. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified.

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Lot size (yards)	Sample size (number of sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE III. Component tests

Component	Characteristic	Requirement reference	FED-STD-191 test method
Base cloth	Material identification	3.2.1	1/
	Weight, oz./sq. yd.	3.2.1	5041
	Weave	3.2.1	Visual
	Color	3.2.1	Visual
	Yarns per inch	3.2.1	5050
	Breaking strength	3.2.1	5100
Coating compound	Composition of silicone rubber	3.2.2	1/

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

4.4.2 End item examination.

4.4.2.1 Yard-by-yard examination. The coated cloth shall be examined for the defects listed below. The required yardage of each roll in the sample shall be examined on one side only; however, the side shall be alternated for every other roll examined. The same yardage shall be given a through-light inspection for pinholes and thinly coated areas. The through-lighting inspection shall be performed in accordance with MIL-STD-1487. The defects found shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The sample unit shall be 1 linear yard. The inspection level shall be I and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0 for major defects and 6.5 for total (major and minor combined) defects. The number of rolls from which the sample is to be selected shall be in accordance with table IV. An approximately equal number of yards shall be examined in each roll sampled.

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Defect	Classification Major	Minor
Any cut, hole, tear, scratch, or abrasion mark that results in baring the base cloth.	X	
Any pinhole.	X	
Any surface scratch or abrasion mark that does not result in baring the base cloth.		X
Any uncoated area.	X	
Any pit, blister, tunnel, or delamination of coating.	X	
Any lump or heavily coated area.		X
Any crease or wrinkle resulting in doubling or adhesion of surfaces that cannot be corrected by manual pressure.	X	
Uneven coating, thin area where coating is missing or noticeably thinner.	X	
Any separation of coating from base cloth.	X	
Any light area or window (resulting from improper distribution of pigment).		X
Any foreign matter, which on removal leaves an uncoated or poorly coated area.	X	
Any foreign matter resulting in unsightly appearance.		X
Edges not straight and uniform.		X
Edges heavily beaded, cannot be rolled in uniform layers.		X
Cloth edges rolled, curled, folded, doubled, scalloped or wavy precluding a flat lay of the cloth.	X	
Any spot, stain, or streak more than 1 inch in combined direction. 1/	X	
Width less than minimum specified.	X	
Any objectionable odor. 2/		X
Color not as specified, off shade, uneven, or mottled.		X
Any tackiness (coating will adhere and not unroll readily).	X	
Deposition of coating not as required, i.e., not balanced as specified.	X	
Not clean.		X

1/ Clearly visible at normal inspection distance (approximately 3 feet).

2/ Odors of chemicals commonly used in coating compounds shall not be regarded as objectionable.

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TABLE IV. Sample size and acceptance criteria

Lot size (yards)	Sample size (rolls)	Maximum number of defects acceptable in sample 2/
Up to 1200 inclusive 1/	3	0
1201 up to and including 3200	5	0
3201 up to and including 10,000	8	0
10,001 up to and including 35,000	13	0
35,001 up to and including 150,000	20	1
150,001 and over	32	2

1/ If a lot contains fewer than three rolls, each roll in the lot shall be examined.

2/ Applicable to length examination only (see 4.4.2.2.1).

4.4.2.2 Length examination.

* 4.4.2.2.1 Examination for length in individual roll. Each individual roll in the sample shall be examined for the defects listed below. If the total number of defects in the sample rolls exceed the maximum number specified in table IV, the lot shall be rejected.

Any roll with a total length of less than 100 yards.
Any roll containing more than three pieces.
Any piece in roll less than 10 yards in length.
Ends of pieces in roll not overlapped, but joined by a seam.
Any roll with a total length more than 2 yards less than that marked on ticket.

4.4.2.2.2 Total yardage in samples. The rolls examined shall be those selected for examination of individual rolls as specified in 4.4.2.2.1. The lot shall be unacceptable if the total of the actual gross lengths of the rolls in the sample is less than the total gross lengths marked on the roll tickets.

* 4.4.3 End item testing. The coated cloth shall be tested for the characteristics listed in table V. The methods of testing specified in FED-STD-191, whenever applicable and as listed in table V, shall be followed. The physical values specified in section 3 apply to the average of determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit for testing shall be 3-1/2 continuous yards full width of the coated cloth. The lot size shall be expressed in units of 1 linear yard and the sample size (number of sample units) shall be as specified below. The lot shall be unacceptable if any unit fails to meet any requirement specified. All test reports shall contain the individual values used in expressing the final result.

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Lot size (yards)	Sample size (number of sample units)
800 or less	2
801 to 22,000	3
22,001 and over	5

4.4.3.1 Standard test conditions. Results of physical tests obtained under testing conditions defined in FED-STD-191, FED-STD-406, or FED-STD-601 will be acceptable except in cases of dispute. In dispute cases, tests shall be conducted with both the specimen and test apparatus under standard conditions as defined in FED-STD-191.

* TABLE V. End item tests

Characteristic	Requirement paragraph	Test method
Weight	3.3	5041
Hydrostatic resistance:		
Initial	3.3	5512
After strength of coating	3.3	5972 and 5512
After abrasion	3.3	4.5.1
After heat aging	3.3	4.5.2
After crease resistance	3.3	4.5.3
Tearing strength	3.3	ASTM D 1424 1/
Breaking strength	3.3	5100
Adhesion of coating	3.3	5970 2/
Stiffness (warp only):		
At room temperature	3.3	5206 3/
At minus 85 ± 2°F	3.3	5206 3/
After hear aging	3.3	5206 3/
Fluid resistance	3.3.1	4.5.4
Aluminum corrosion	3.3.2	4.5.5

1/ See 2.2

2/ Except that a silicone adhesive shall be employed and allowed to age 4 days between glass plates and 3 additional days at room temperature after removal from plates (see 6.5).

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3/ Except that the value shall be only the full reading in inches and not more than 15 seconds shall elapse between the start of test and the final reading.

4.4.4 Preparation for delivery inspection. The inspection shall be in accordance with the quality assurance provisions of PPP-P-1136.

4.5 Methods of inspection.

4.5.1 Hydrostatic resistance after abrasion. The abrasion resistance procedure shall be conducted as specified in Method 5302 of FED-STD-191 with the following exceptions:

a. A solid rubber diaphragm 0.030 ± 0.010 inch thick with a non-metallic contact shall be used.

b. Multidirectional abrasion shall be employed and 50 double cycles shall constitute the test on each specimen.

c. The abradant shall be fine emery cloth and ten specimens shall be abraded, five on one side of the coated cloth and five on the reverse side of the coated cloth.

The hydrostatic resistance shall be conducted as specified in Method 5512 of FED-STD-191. The average of ten hydrostatic readings shall constitute a test, five on each side of the coated cloth with the abraded side up and with the abraded portion of the test specimen in the center of the test area.

4.5.2 Hydrostatic resistance after heat aging. A 10 by 12 inch specimen of the coated cloth shall be exposed to a temperature of 200 ± 2°F for 8 days in a circulating air oven. After exposure the specimen shall be examined for evidence of tackiness, flaking, brittleness, stiffness and then tested for hydrostatic resistance as specified in Method 5512 of FED-STD-191.

4.5.3 Hydrostatic resistance after crease resistance. A 6 by 6 inch sample of the coated cloth shall be exposed to a temperature of minus 85 ± 2°F for 20 hours. A steel roller, approximately 5-1/2 inches in diameter, approximately 2 inches wide, with a suitable handle guide and having a total weight of 10 pounds shall also be conditioned at this temperature and time period. Without being removed from the specified temperature conditions, the specimen shall be folded in the warp direction. Without opening the fold, the specimen shall then be folded 90° in the filling direction. The test specimen shall be placed on a flat smooth surface and the center of the steel roller shall be run once across the folds. The specimen shall then be removed from the conditioning chamber and examined for cracking, flaking, and separation of the coating from the base fabric and then tested for hydrostatic resistance in the folded area as specified in Method 5512 of FED-STD-191.

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4.5.4 Fluid resistance test. Four 2 by 3-inch specimens shall be cut from the coated cloth. The four specimens shall be immersed in 100 mL of each of the following fluids (one specimen to each fluid) for a period of 30 ± 1 minutes at 77 ± 4°F:

a. Type I standard test fluid per TT-S-735.

b. Grade JP-4 fuel per MIL-T-5624.

c. Hydraulic fluid per MIL-H-5606.

d. Grade 1005 lubricating oil per MIL-L-6081.

At the end of the immersion period the specimens shall be removed from the fluids, allowed to drain and dry for 48 hours and then examined for evidence of tackiness, peeling, flaking, or stiffening.

4.5.5 Aluminum corrosion test. Both of the following corrosion tests shall be conducted:

a. Each of the three 2024-T3 aluminum strips conforming to QQ-A-250/4 approximately 6 by 1 by 1/8 inches in dimension shall have a 1 inch strip of coated cloth wrapped around the strip for a distance of 4 inches on each strip. Three additional strips, as specified above, shall be wrapped with filter paper in lieu of the coated cloth and shall be considered the control specimens. All specimens shall then be placed in an atmosphere of 95 ± 2 percent relative humidity and 95 ± 2°F temperature for 150 ± 2 hours. After exposure the specimens shall be removed from the conditioning chamber, unwrapped, and examined for visible pitting of the test specimens in excess of that observed on the control specimens.

b. A 6 inch by 6 inch specimen of the coated cloth shall be placed in a container filled with enough distilled water to cover the material. After soaking for 22 ± 2 hours at room temperature, the liquid shall be poured into a clean evaporating dish containing a piece of non-clad 2024-T3 aluminum alloy conforming to specification QQ-A-250/4. The liquid shall be allowed to evaporate to dryness at a temperature not exceeding 158°F. A control sample shall be prepared by immersing a piece of aluminum as specified above in an equivalent amount of distilled water and evaporating the water in the same manner. Both samples shall then be sponged, rinsed and dried, and examined for pitting. Visible pitting, but not discoloration of the test panel in excess of that observed on the control specimens shall be considered as evidence of corrosion.

5. PREPARATION FOR DELIVERY

5.1 Put-up and packaging. Packaging shall be level A or C as specified (see 6.2).

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5.1.1 Levels A and C. The coated cloth put up as specified, shall be packaged in accordance with the applicable requirements of PPP-P-1136.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Levels A, B, and C. The coated cloth shall be packed in accordance with the applicable requirements of PPP-P-1136.

5.3 Marking. In addition to any special marking required by the contract or purchase order, interior packages and shipping containers shall be marked in accordance with PPP-P-1136.

6. NOTES

6.1 Intended use. The coated cloth covered by this document is intended for use in aerodynamic balance seals.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this document.
b. When a first article is required (see 3.1.2, 4.3 and 6.4).
c. Color required (see 3.4).
d. Width required (see 3.5).
e. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to standard sample, address the contracting activity issuing the invitation for bids (see 3.1.1).

6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample consisting of 3-1/2 yards of the finished coated cloth. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.5 Adhesive sealant. Dow Corning adhesive sealant "Silastic 732 RTV' has proven satisfactory for the adhesion of coating test.

6.6 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.2).

6.7 Changes from previous issue. The margins of this document are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording.

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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